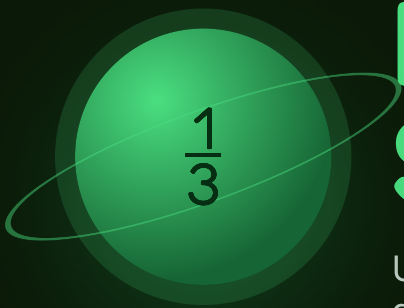


Fraction Sums

Unlike denominators, and a common one you find rather than guess.

THE BIG QUESTION

Why can't you add 1/2 and 1/3 by adding the tops and adding the bottoms?

HANDS-ON PROJECT

Double the Recipe

Take a real recipe, scale it up, and cook it. The measuring cups do not lie.

FLUENCY GAME

Denominator Hunt

Two fractions drawn at random. First to name the smallest common denominator wins.

AXIOM · MISSION COMMANDER, NINE TOURS

“You cannot add halves to thirds any more than you can add metres to gallons. Change the units first, then add.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

NAME ______________

DATE ______________

This is not a test. Nothing here counts for a grade. Write fractions as a/b — for example, three quarters is 3/4.

STRAND A EQUIVALENTS	A1 1/2 = ?/6	A2 2/3 = ?/12
STRAND B SAME DENOMINATOR	B1 3/8 + 2/8	B2 5/6 – 1/6
STRAND C COMMON DENOMINATOR	C1 Smallest common denominator of 4 and 6	C2 Of 3 and 5
STRAND D UNLIKE DENOMINATORS	D1 1/2 + 1/3	D2 3/4 – 1/6
STRAND E MIXED NUMBERS	E1 Write 7/4 as a mixed number	E2 1 1/2 + 2 1/4
STRAND F ESTIMATING	F1 Is 5/8 + 1/3 more or less than 1?	F2 Someone says 1/2 + 1/3 = 2/5. Why is that wrong?

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the five-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Equivalents	A1 A2	Week 1 · lesson 1.1	___ / 2	Day 1.1. Equivalent fractions are the engine of the whole mission, so check this is solid rather than lucky.
B · Same denominator	B1 B2	Week 1 · lesson 1.2	___ / 2	Day 1.2. This is Year One material.
C · Common denominator	C1 C2	Week 2 · lessons 2.1 and 2.2	___ / 2	Compress Week 2 to three days. Keep 2.3 — the multiples ladder is worth the time.
D · Unlike denominators	D1 D2	Week 3 · all four lessons	___ / 2	Compress Week 3 to three days. Do not remove it entirely — this is the mission.
E · Mixed numbers	E1 E2	Week 4 · lessons 4.1 to 4.3	___ / 2	Days 4.1 and 4.2. Keep 4.3 — borrowing across a whole number is its own skill.
F · Estimating	F1 F2	Week 2 · lesson 2.4	___ / 2	Nothing. F2 is the Big Question in miniature — teach it even at 2/2.

ONE NOTE BEFORE YOU START

The answer 2/5 to item D1 is the most informative wrong answer in the whole year. It means the student is treating a fraction as two separate numbers rather than one quantity. If you see it, spend Week 1 on the area diagrams rather than the procedure — no amount of practice at finding common denominators fixes that misconception.

MISSION 05 · WEEK 1 · MONDAY · LESSON 1.1

Same Amount, New Name

Multiply top and bottom by the same number and nothing changes.

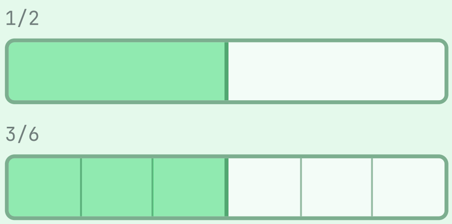
NAME ___________

DATE ___________

Warm-Up FILL IN THE MISSING TOP

1/2 = ?/4	1/2 = ?/6	1/3 = ?/6	1/4 = ?/8	2/3 = ?/9	3/4 = ?/12

1/2 AND 3/6 ARE THE SAME AMOUNT



The shaded amount is identical. All that changed is how many pieces the bar was cut into — and cutting each piece into three makes three times as many pieces, three times as many of them shaded.

That is why multiplying top and bottom by 3 is allowed: **1/2 = 3/6**. You are renaming the amount, not changing it.

Core

1. 2/3 = ?/12

2. 3/5 = ?/20

3. Simplify 8/12

4. Simplify 15/20

5. Write three different fractions that all equal 1/2. Say what you did to get each one.

Challenge NOT GRADED

C1. Simplify 36/48 all the way down. How do you know you have finished?

C2. Why does adding 1 to the top and 1 to the bottom of 1/2 not give an equivalent fraction? Draw it if that helps.

AXIOM: Multiplying is allowed. Adding is not. Top and bottom must be scaled by the same factor, or you have a different fraction.

When the Bottoms Match

Add the tops. The bottom is the unit, and it stays put.

NAME ___________

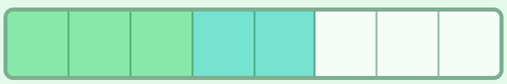
DATE ___________

Warm-Up

1/8 + 3/8	3/8 + 2/8	5/6 - 1/6	2/5 + 2/5	7/10 - 3/10	3/4 + 1/4

THE DENOMINATOR NAMES THE UNIT

3/8 + 2/8 = 5/8



Three eighths plus two eighths is five eighths, the same way three apples plus two apples is five apples. The word "eighths" is the unit, and it does not change just because you added some more of them.

This is exactly why you cannot add 1/2 to 1/3 directly. Halves and thirds are different units, and there is no single word for the answer until you make them the same.

Core

SIMPLIFY EVERY ANSWER

1. 3/8 + 2/8

2. 2/6 + 2/6

3. 9/10 - 4/10

4. 5/12 + 7/12

5. In item 4 the answer is a whole number. Explain what happened to the twelfths.

Challenge

C1. 7/8 + 5/8. The answer is more than one whole — write it as a mixed number.

C2. Somebody works out 3/8 + 2/8 and gets 5/16. Which part did they add that they should have left alone?

NAME ___________

DATE ___________

Compare Without Adding

Benchmark against one half. Faster than a common denominator.

Warm-Up

MORE OR LESS THAN A HALF?

3/8	5/8	2/5	7/12	4/9	11/20
___	___	___	___	___	___

THREE WAYS TO COMPARE, FASTEST FIRST

AGAINST A HALF

3/8 is under a half, 5/9 is over. Done — no arithmetic needed.

SAME TOP

3/5 beats 3/8, because fifths are bigger pieces than eighths.

COMMON DENOMINATOR

Always works, slowest. Save it for when the first two do not settle it.

A half is the useful landmark because you can see it at a glance: double the top and compare it to the bottom.

Core

SAY WHICH METHOD YOU USED

1. Larger: 3/8 or 5/9

2. Larger: 3/5 or 3/8

3. Larger: 2/3 or 5/8

4. Larger: 7/12 or 5/8

5. Put in order, smallest first: 2/5   1/2   5/8   3/10

Challenge

C1. Which is larger, 7/8 or 8/9? Both are just under one whole, so the half trick will not help.

C2. Write a fraction between 1/2 and 5/8. How did you find it?

NAME ___________

DATE ___________

Why 2/5 Is Wrong

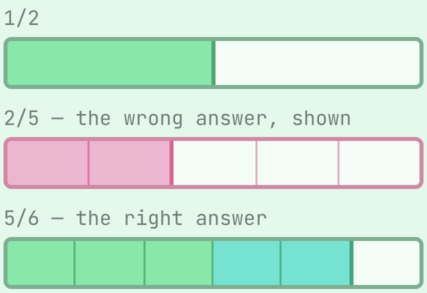
The most common fraction error in the world, taken apart with a diagram.

Warm-Up

ESTIMATE: MORE OR LESS THAN 1?

1/2 + 1/3	3/4 + 1/2	1/4 + 1/3	5/8 + 1/2	1/5 + 1/4	2/3 + 2/3
______	______	______	______	______	______

1/2 + 1/3 · THE ANSWER MUST BE BIGGER THAN A HALF



You started with a half and added something to it. The answer **has** to be more than a half.

But 2/5 is *less* than a half. So it is wrong before you check any arithmetic — and that check takes one second.

Adding the bottoms makes the pieces smaller, which is the opposite of what adding should do.

Core

ESTIMATE BEFORE YOU COMPUTE

1. Is 1/2 + 1/3 more or less than 1?

2. Is 5/8 + 1/3 more or less than 1?

3. Is 3/4 + 1/2 more or less than 1?

4. Is 1/5 + 1/4 more or less than 1/2?

5. Somebody says 1/4 + 1/4 = 2/8. Use the “more than what you started with” test to show it is wrong.

Challenge

C1. Adding the tops and bottoms of 1/2 and 1/2 gives 2/4, which equals 1/2. Why does this look like it worked, and what went wrong anyway?

C2. Can adding two fractions ever give an answer smaller than both of them? Say why or why not.

NAME ___________

DATE ___________

Denominator Hunt

First to name the smallest common denominator takes the round.

How to play: write the numbers 2 to 12 on slips of paper. Draw two. Both players race to name the smallest number both will divide into. Say it out loud; first correct answer wins. If the two numbers share a factor, the answer is *not* their product — that is where the round is won.

Warm-Up · list the multiples

First six multiples of 4

....

First six multiples of 6

....

Circle the first number that appears in both lists. That is the smallest common denominator for quarters and sixths — and it is 12, not 24.

Play eight rounds

ROUND	NUMBERS DRAWN	SMALLEST COMMON	THEIR PRODUCT	SAME?
1				
2				
3				
4				
5				
6				
7				
8				

Challenge

C1. Look down your table. When is the smallest common denominator the same as the product? What do those pairs have in common?

....

C2. Which pair from 2 to 12 has the largest smallest-common-denominator? Find it and say why.

....

Weeks 2–5

Every day below has five graded sets waiting in Practice Bay.

QUIZ · end of week 3
TEST · end of week 5

Week 2 · Finding the Common Denominator

Multiples, not guesses. The smallest common denominator is rarely the product.

MON 2.1 — **List the multiples**
First shared one wins.

TUE 2.2 — **Rename both fractions**
Scale each one to the new bottom.

WED 2.3 — **The multiples ladder**
Why 4 and 6 give 12, not 24.

THU 2.4 — **Estimate the answer**
More or less than one, before any arithmetic.

FRI — **Denominator Hunt**
Page 8's game, played for speed.

Week 3 · Adding Unlike Fractions QUIZ FRIDAY

The mission's core week. Rename, add, simplify — and estimate first, every time.

MON 3.1 — **Halves and thirds**
The classic pair, done properly.

TUE 3.2 — **Quarters and sixths**
Where the shared factor saves work.

WED 3.3 — **Subtracting unlike**
Same method, minus sign.

THU 3.4 — **Simplify the answer**
Not finished until it is in lowest terms.

FRI — **Mid-unit quiz**
Page 10. Eight items, 85%.

Week 4 · Mixed Numbers

Wholes and parts together, including borrowing from the whole number.

MON 4.1 — **Improper and mixed**
Convert both ways.

TUE 4.2 — **Adding mixed numbers**
Wholes first, then the parts.

WED 4.3 — **Borrowing from a whole**
3 becomes 2 and 4/4.

THU 4.4 — **Recipe arithmetic**
Real measurements, doubled and halved.

FRI — **Recipe chosen**
Pick it, write out the doubled amounts.

Week 5 · Proof UNIT TEST

The recipe gets cooked. Measuring cups are unforgiving, which is the point.

MON 5.1 — **Scale every line**
Doubled, then halved.

TUE 5.2 — **Cook it**
Wrong arithmetic tastes wrong.

WED 5.3 — **Mixed review**
All five weeks together.

THU — **Error journal sweep**
Fix only what repeats.

FRI — **Mission 05 test**
Pages 11–12.

MISSION 05 · END OF WEEK 3

Mid-Unit Quiz

NAME ______________

SCORE ______ / 8

Eight items from Weeks 1 to 3. 85% to keep flying — that's 7 out of 8.

1. 1/2 = ?/6 ....

2. 3/8 + 2/8 ....

3. Smallest common denominator of 4 and 6 ....

4. 1/2 + 1/3 ....

5. 3/4 + 1/6 ....

6. 5/6 – 1/4 ....

7. Simplify 8/12 ....

8. Somebody says 1/2 + 1/3 = 2/5. Explain why that answer cannot be right, without working out the real one.

....

....

Marking note: item 8 wants the size argument. Full marks for "you added something to a half, so the answer must be bigger than a half, and 2/5 is smaller."

ANSWERS ON PAGE 15

Unit Test · Part 1

NAME ______________

DATE ______________

Items 1–8. No time limit. Simplify every answer, and estimate first.

1. 2/3 = ?/12 ....

2. 5/12 + 7/12 ....

3. 1/2 + 1/3 ....

4. 2/3 + 3/4 ....

5. 3/4 – 1/6 ....

6. Write 7/4 as a mixed number ....

7. 1 1/2 + 2 1/4 ....

8. Larger: 7/12 or 5/8 ....

CONTINUES ON PAGE 12

Unit Test · Part 2

Items 9–12 and the Big Question.

SCORE ______ / 12

BAND ______________

9. 3 – 1 3/4

10. 2/3 + 1/4 + 1/2

11. A recipe needs 3/4 cup of flour and 2/3 cup of sugar. How much dry ingredient altogether? Is that more or less than 1 1/2 cups?

12. You double a recipe that called for 1 1/3 cups of milk. How much do you need now? Write it as a mixed number.

EXPLAIN YOUR THINKING · THE BIG QUESTION

Why can't you add 1/2 and 1/3 by adding the tops and adding the bottoms?

Draw a diagram or use the “bigger than a half” argument. Scored on reasoning, not neatness.

11–12
Gold band. The Common Ground Badge is earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 3/6 | A2 · 8/12 | B1 · 5/8 | B2 · 4/6 or 2/3

C1 · 12 | C2 · 15 | D1 · 5/6 | D2 · 7/12

E1 · 1 3/4 | E2 · 3 3/4 | F1 · less than 1 | F2 · see below

F2 · adding the bottoms makes the pieces smaller, so the answer comes out less than the half you started with. An answer of 2/5 on D1 is the diagnostic — see the note on the skip chart.

Lesson 1.1 · Same Amount, New Name

Warm-Up · 2/4 | 3/6 | 2/6 | 2/8 | 6/9 | 9/12

Core · 1 · 8/12 | 2 · 12/20 | 3 · 2/3 | 4 · 3/4

Item 5 · any three, e.g. 2/4, 3/6, 50/100 — the reason matters more than the choice. C1 · 3/4; you are finished when top and bottom share no factor but 1. C2 · because 2/3 is not 1/2 — adding changes the ratio, while multiplying preserves it.

Lesson 1.2 · When the Bottoms Match

Warm-Up · 4/8 or 1/2 | 5/8 | 4/6 or 2/3 | 4/5 | 4/10 or 2/5 | 1

Core · 1 · 5/8 | 2 · 4/6 = 2/3 | 3 · 5/10 = 1/2 | 4 · 12/12 = 1

Item 5 · twelve twelfths make one whole, so the fraction disappears. C1 · 12/8 = 1 1/2. C2 · they added the denominators; eighths plus eighths are still eighths.

TEACHER PAGE · ANSWER KEY

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · Compare Without Adding

Warm-Up · less | more | less | more | less | more

Core · 1 · 5/9 | 2 · 3/5 | 3 · 2/3 | 4 · 5/8 | 5 · 3/10, 2/5, 1/2, 5/8

C1 · 8/9, because 1/9 missing is less than 1/8 missing — comparing what is left over is the trick. C2 · e.g. 9/16, found by using the common denominator 16.

Lesson 1.4 · Why 2/5 Is Wrong

Warm-Up · less | more | less | more | less | more

Core · 1 · less | 2 · less | 3 · more | 4 · less

Item 5 · you added a quarter to a quarter, so the answer must be bigger than a quarter; 2/8 *is* a quarter, so nothing was added. C1 · it worked by coincidence — 2/4 does equal 1/2, but only because both fractions were identical halves; try 1/2 + 1/3 and the method collapses. C2 · no, never, as long as both fractions are positive: adding a positive amount always increases the total.

Friday · Denominator Hunt

Warm-Up · 4, 8, 12, 16, 20, 24 | 6, 12, 18, 24, 30, 36 → first shared is 12

C1 · the product only works when the two numbers share no factor — 3 and 5 give 15, but 4 and 6 give 12 rather than 24. C2 · 11 and 12, giving 132, since 11 is prime and shares nothing with 12.

TEACHER PAGE · ANSWER KEY

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 3/6

2 · 5/8

3 · 12

4 · 5/6

5 · 11/12

6 · 7/12

7 · 2/3

8 · size argument

Unit test

1 · 8/12

2 · 1

3 · 5/6

4 · 17/12 = 1 5/12

5 · 7/12

6 · 1 3/4

7 · 3 3/4

8 · 5/8

9 · 1 1/4

10 · 17/12 = 1 5/12

Items 11, 12 and the Big Question

11 · 17/12 cups, or 1 5/12 — less than 1 1/2, since 5/12 is under a half. **12** · 2 2/3 cups. **Big Question** · full marks for either route: a diagram showing that halves and thirds are different-sized pieces so the counts cannot be combined until they match, or the size argument that 2/5 is less than the 1/2 you started with. An answer that only recites “find a common denominator” without saying why scores half — the procedure is not the reason.

Error journal · Mission 05

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 5, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY